Understanding Student Performance: R Analysis

Final Project, Scenario 2 | LEAP Data Analytics and Statistics with Applications

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October 2026

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0. Setup

```
needed_packages <- c("ggplot2", "dplyr", "broom")

missing_packages <- needed_packages[!(needed_packages %in% rownames(installed.packages()))]
if (length(missing_packages) > 0) {
  install.packages(missing_packages)
}

library(ggplot2)
library(dplyr)
library(broom)

options(digits = 4)

# Output folders for figures and tables used in the report and slides.
dir.create("figures", showWarnings = FALSE)
dir.create("results", showWarnings = FALSE)

# Helper: save the most recent ggplot with consistent dimensions.
save_fig <- function(name, width = 6, height = 4) {
  ggsave(file.path("figures", paste0(name, ".png")),
         width = width, height = height, dpi = 200, bg = "white")
}
```

```r
# A single, quiet theme for all figures.
theme_set(theme_minimal(base_size = 11))
```

Question 1. Understand and prepare the data

```r
local_file <- "student-mat.csv"

# Download from UCI only if the local copy is missing.
if (!file.exists(local_file)) {
  zip_url <- "https://archive.ics.uci.edu/static/public/320/student+performance.zip"
  zip_file <- tempfile(fileext = ".zip")
  extract_dir <- tempfile("student_data_")
  dir.create(extract_dir)

  download.file(zip_url, zip_file, mode = "wb")

  zip_contents <- unzip(zip_file, list = TRUE)$Name
  math_name <- zip_contents[grepl("student-mat\\.csv$", zip_contents)]

  if (length(math_name) != 1) {
    stop("Could not identify student-mat.csv inside the downloaded ZIP file.")
  }

  unzip(zip_file, files = math_name, exdir = extract_dir)
  file.copy(file.path(extract_dir, math_name), local_file, overwrite = TRUE)
}

# The UCI file is semicolon separated.
students <- read.csv2(local_file, stringsAsFactors = FALSE)

head(students)
```

```
##   school sex age address famsize Pstatus Medu Fedu     Mjob     Fjob     reason
## 1     GP   F  18       U     GT3       A    4    4  at_home  teacher     course
## 2     GP   F  17       U     GT3       T    1    1  at_home    other     course
## 3     GP   F  15       U     LE3       T    1    1  at_home    other      other
## 4     GP   F  15       U     GT3       T    4    2   health services       home
## 5     GP   F  16       U     GT3       T    3    3    other    other       home
## 6     GP   M  16       U     LE3       T    4    3 services    other reputation
##   guardian traveltime studytime failures schoolsup famsup paid activities
## 1   mother          2         2        0       yes     no   no         no
## 2   father          1         2        0        no    yes   no         no
## 3   mother          1         2        3       yes     no  yes         no
## 4   mother          1         3        0        no    yes  yes        yes
## 5   father          1         2        0        no    yes  yes         no
## 6   mother          1         2        0        no    yes  yes        yes
##   nursery higher internet romantic famrel freetime goout Dalc Walc health
## 1     yes    yes       no       no      4        3     4    1    1      3
## 2      no    yes      yes       no      5        3     3    1    1      3
## 3     yes    yes      yes       no      4        3     2    2    3      3
## 4     yes    yes      yes      yes      3        2     2    1    1      5
```

```
## 5     yes    yes       no       no      4        3     2    1    2      5
## 6     yes    yes      yes       no      5        4     2    1    2      5
##   absences G1 G2 G3
## 1        6  5  6  6
## 2        4  5  5  6
## 3       10  7  8 10
## 4        2 15 14 15
## 5        4  6 10 10
## 6       10 15 15 15
```

```
dim(students)          # 395 rows (students), 33 columns
```

```
## [1] 395  33
```

```
names(students)
```

```
##  [1] "school"     "sex"        "age"        "address"    "famsize"
##  [6] "Pstatus"    "Medu"       "Fedu"       "Mjob"       "Fjob"
## [11] "reason"     "guardian"   "traveltime" "studytime"  "failures"
## [16] "schoolsup"  "famsup"     "paid"       "activities" "nursery"
## [21] "higher"     "internet"   "romantic"   "famrel"     "freetime"
## [26] "goout"      "Dalc"       "Walc"       "health"     "absences"
## [31] "G1"         "G2"         "G3"
```

```
str(students)
```

```
## 'data.frame':    395 obs. of  33 variables:
##  $ school    : chr  "GP" "GP" "GP" "GP" ...
##  $ sex       : chr  "F" "F" "F" "F" ...
##  $ age       : int  18 17 15 15 16 16 16 17 15 15 ...
##  $ address   : chr  "U" "U" "U" "U" ...
##  $ famsize   : chr  "GT3" "GT3" "LE3" "GT3" ...
##  $ Pstatus   : chr  "A" "T" "T" "T" ...
##  $ Medu      : int  4 1 1 4 3 4 2 4 3 3 ...
##  $ Fedu      : int  4 1 1 2 3 3 2 4 2 4 ...
##  $ Mjob      : chr  "at_home" "at_home" "at_home" "health" ...
##  $ Fjob      : chr  "teacher" "other" "other" "services" ...
##  $ reason    : chr  "course" "course" "other" "home" ...
##  $ guardian  : chr  "mother" "father" "mother" "mother" ...
##  $ traveltime: int  2 1 1 1 1 1 1 2 1 1 ...
##  $ studytime : int  2 2 2 3 2 2 2 2 2 2 ...
##  $ failures  : int  0 0 3 0 0 0 0 0 0 0 ...
##  $ schoolsup : chr  "yes" "no" "yes" "no" ...
##  $ famsup    : chr  "no" "yes" "no" "yes" ...
##  $ paid      : chr  "no" "no" "yes" "yes" ...
##  $ activities: chr  "no" "no" "no" "yes" ...
##  $ nursery   : chr  "yes" "no" "yes" "yes" ...
##  $ higher    : chr  "yes" "yes" "yes" "yes" ...
##  $ internet  : chr  "no" "yes" "yes" "yes" ...
##  $ romantic  : chr  "no" "no" "no" "yes" ...
##  $ famrel    : int  4 5 4 3 4 5 4 4 4 5 ...
##  $ freetime  : int  3 3 3 2 3 4 4 1 2 5 ...
```

```
##  $ goout     : int  4 3 2 2 2 2 4 4 2 1 ...
##  $ Dalc      : int  1 1 2 1 1 1 1 1 1 1 ...
##  $ Walc      : int  1 1 3 1 2 2 1 1 1 1 ...
##  $ health    : int  3 3 3 5 5 5 3 1 1 5 ...
##  $ absences  : int  6 4 10 2 4 10 0 6 0 0 ...
##  $ G1        : int  5 5 7 15 6 15 12 6 16 14 ...
##  $ G2        : int  6 5 8 14 10 15 12 5 18 15 ...
##  $ G3        : int  6 6 10 15 10 15 11 6 19 15 ...
```

```
# Data-quality checks.
colSums(is.na(students))
```

```
##     school        sex        age    address    famsize    Pstatus       Medu
##          0          0          0          0          0          0          0
##       Fedu       Mjob       Fjob     reason   guardian traveltime  studytime
##          0          0          0          0          0          0          0
##   failures  schoolsup     famsup       paid activities    nursery     higher
##          0          0          0          0          0          0          0
##   internet   romantic     famrel   freetime      goout       Dalc       Walc
##          0          0          0          0          0          0          0
##     health   absences         G1         G2         G3
##          0          0          0          0          0
```

```
sum(is.na(students))       # 0 missing values
```

```
## [1] 0
```

```
sum(duplicated(students))  # 0 duplicated rows
```

```
## [1] 0
```

```
summary(students)
```

```
##        school              sex                 age             address         famsize
##  Length   :395   Length   :395   Min.   :15.0   Length   :395   Length   :395
##  N.unique :  2   N.unique :  2   1st Qu.:16.0   N.unique :  2   N.unique :  2
##  N.blank  :  0   N.blank  :  0   Median :17.0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  2   Min.nchar:  1   Mean   :16.7   Min.nchar:  1   Min.nchar:  3
##  Max.nchar:  2   Max.nchar:  1   3rd Qu.:18.0   Max.nchar:  1   Max.nchar:  3
##                                  Max.   :22.0
##       Pstatus          Medu            Fedu            Mjob            Fjob
##  Length   :395   Min.   :0.00   Min.   :0.00   Length   :395   Length   :395
##  N.unique :  2   1st Qu.:2.00   1st Qu.:2.00   N.unique :  5   N.unique :  5
##  N.blank  :  0   Median :3.00   Median :2.00   N.blank  :  0   N.blank  :  0
##  Min.nchar:  1   Mean   :2.75   Mean   :2.52   Min.nchar:  5   Min.nchar:  5
##  Max.nchar:  1   3rd Qu.:4.00   3rd Qu.:3.00   Max.nchar:  8   Max.nchar:  8
##                  Max.   :4.00   Max.   :4.00
##        reason          guardian     traveltime     studytime       failures
##  Length   :395   Length   :395   Min.   :1.00   Min.   :1.00   Min.   :0.000
##  N.unique :  4   N.unique :  3   1st Qu.:1.00   1st Qu.:1.00   1st Qu.:0.000
##  N.blank  :  0   N.blank  :  0   Median :1.00   Median :2.00   Median :0.000
```

```
##  Min.nchar:  4   Min.nchar:  5   Mean   :1.45   Mean   :2.04   Mean   :0.334
##  Max.nchar: 10   Max.nchar:  6   3rd Qu.:2.00   3rd Qu.:2.00   3rd Qu.:0.000
##                                  Max.   :4.00   Max.   :4.00   Max.   :3.000
##      schoolsup          famsup           paid         activities
##  Length   :395   Length   :395   Length   :395   Length   :395
##  N.unique :  2   N.unique :  2   N.unique :  2   N.unique :  2
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0
##  Min.nchar:  2   Min.nchar:  2   Min.nchar:  2   Min.nchar:  2
##  Max.nchar:  3   Max.nchar:  3   Max.nchar:  3   Max.nchar:  3
##
##       nursery          higher         internet        romantic       famrel
##  Length   :395   Length   :395   Length   :395   Length   :395   Min.   :1.00
##  N.unique :  2   N.unique :  2   N.unique :  2   N.unique :  2   1st Qu.:4.00
##  N.blank  :  0   N.blank  :  0   N.blank  :  0   N.blank  :  0   Median :4.00
##  Min.nchar:  2   Min.nchar:  2   Min.nchar:  2   Min.nchar:  2   Mean   :3.94
##  Max.nchar:  3   Max.nchar:  3   Max.nchar:  3   Max.nchar:  3   3rd Qu.:5.00
##                                                                  Max.   :5.00
##     freetime        goout           Dalc           Walc          health
##  Min.   :1.00   Min.   :1.00   Min.   :1.00   Min.   :1.00   Min.   :1.00
##  1st Qu.:3.00   1st Qu.:2.00   1st Qu.:1.00   1st Qu.:1.00   1st Qu.:3.00
##  Median :3.00   Median :3.00   Median :1.00   Median :2.00   Median :4.00
##  Mean   :3.24   Mean   :3.11   Mean   :1.48   Mean   :2.29   Mean   :3.55
##  3rd Qu.:4.00   3rd Qu.:4.00   3rd Qu.:2.00   3rd Qu.:3.00   3rd Qu.:5.00
##  Max.   :5.00   Max.   :5.00   Max.   :5.00   Max.   :5.00   Max.   :5.00
##     absences           G1              G2              G3
##  Min.   : 0.00   Min.   : 3.0   Min.   : 0.0   Min.   : 0.0
##  1st Qu.: 0.00   1st Qu.: 8.0   1st Qu.: 9.0   1st Qu.: 8.0
##  Median : 4.00   Median :11.0   Median :11.0   Median :11.0
##  Mean   : 5.71   Mean   :10.9   Mean   :10.7   Mean   :10.4
##  3rd Qu.: 8.00   3rd Qu.:13.0   3rd Qu.:13.0   3rd Qu.:14.0
##  Max.   :75.00   Max.   :19.0   Max.   :19.0   Max.   :20.0
```

```r
# Convert categorical (nominal and binary) variables to factors.
factor_variables <- c(
  "school", "sex", "address", "famsize", "Pstatus",
  "Mjob", "Fjob", "reason", "guardian", "schoolsup", "famsup",
  "paid", "activities", "nursery", "higher", "internet", "romantic"
)
students[factor_variables] <- lapply(students[factor_variables], factor)

# Ordered variables (Medu, Fedu, traveltime, studytime, famrel, freetime,
# goout, Dalc, Walc, health) are kept as integer codes so that they can enter
# correlations and regressions as approximately linear scores. This is a
# simplifying assumption that is discussed in the report.
# failures is a count that is top-coded: the value 3 means "3 or more".

# Binary outcome: Pass if G3 >= 10.
students <- students %>%
  mutate(pass = factor(ifelse(G3 >= 10, "Pass", "Fail"),
                       levels = c("Fail", "Pass")))
table(students$pass)
```

```
##
## Fail Pass
```

```
##  130  265
```

Problematic values: final grade of zero

```
# A Portuguese grade of 0 out of 20 is very unusual for a student who sat the
# exam. Check how many students have G3 == 0 and how they look on other
# variables.
zero_g3 <- students$G3 == 0
sum(zero_g3)                                    # 38 students
```

```
## [1] 38
```

```
table(students$absences[zero_g3])               # all 38 have 0 absences
```

```
##
##  0
## 38
```

```
summary(students$G1[zero_g3])                   # earlier grades are non-zero
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    4.00    6.00    7.00    7.53    9.00   12.00
```

```
sum(students$G2[zero_g3] == 0)                  # 13 already had G2 == 0
```

```
## [1] 13
```

```
# Interpretation: these are most plausibly students who did not sit the final
# assessment (dropped out or did not attend), with absences not recorded,
# rather than students who sat the exam and scored zero. We keep them in the
# main analysis, because they are real fails, but we test in Question 5
# whether the conclusions depend on them.

zero_summary <- students %>%
  mutate(group = ifelse(zero_g3, "G3 = 0", "G3 > 0")) %>%
  group_by(group) %>%
  summarise(n = n(),
            mean_absences = mean(absences),
            share_zero_absences = mean(absences == 0),
            mean_G1 = mean(G1),
            mean_failures = mean(failures))
print(zero_summary)
```

```
## # A tibble: 2 x 6
##   group      n mean_absences share_zero_absences mean_G1 mean_failures
##   <chr>  <int>         <dbl>               <dbl>   <dbl>         <dbl>
## 1 G3 = 0    38          0                  1        7.53         0.921
## 2 G3 > 0   357          6.32               0.216   11.3          0.272
```

```r
write.csv(zero_summary, "results/zero_grade_check.csv", row.names = FALSE)
```

Question 2. Explore the data

```r
describe_numeric <- function(x) {
  c(
    n = sum(!is.na(x)),
    mean = mean(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    sd = sd(x, na.rm = TRUE),
    min = min(x, na.rm = TRUE),
    q1 = unname(quantile(x, 0.25, na.rm = TRUE)),
    q3 = unname(quantile(x, 0.75, na.rm = TRUE)),
    max = max(x, na.rm = TRUE),
    iqr = IQR(x, na.rm = TRUE)
  )
}

desc_vars <- c("G3", "age", "absences", "failures", "studytime", "Medu", "goout")
descriptives <- as.data.frame(t(sapply(students[desc_vars], describe_numeric)))
descriptives$variable <- rownames(descriptives)
print(round(descriptives[, -ncol(descriptives)], 2))
```

```
##             n  mean median   sd min q1 q3 max iqr
## G3        395 10.42     11 4.58   0  8 14  20   6
## age       395 16.70     17 1.28  15 16 18  22   2
## absences  395  5.71      4 8.00   0  0  8  75   8
## failures  395  0.33      0 0.74   0  0  0   3   0
## studytime 395  2.04      2 0.84   1  1  2   4   1
## Medu      395  2.75      3 1.09   0  2  4   4   2
## goout     395  3.11      3 1.11   1  2  4   5   2
```

```r
write.csv(descriptives, "results/descriptives.csv", row.names = FALSE)

# G3 without the zero group, for comparison.
describe_numeric(students$G3[!zero_g3])
```

```
##       n    mean  median      sd     min      q1      q3     max     iqr
## 357.000  11.524  11.000   3.228   4.000   9.000  14.000  20.000   5.000
```

```r
# Frequency tables.
table(students$pass)
```

```
##
## Fail Pass
##  130  265
```

```r
round(prop.table(table(students$pass)) * 100, 1)
```

```
##
## Fail Pass
## 32.9 67.1
```

```r
table(students$sex)
```

```
##
##   F   M
## 208 187
```

```r
table(students$higher)
```

```
##
##  no yes
##  20 375
```

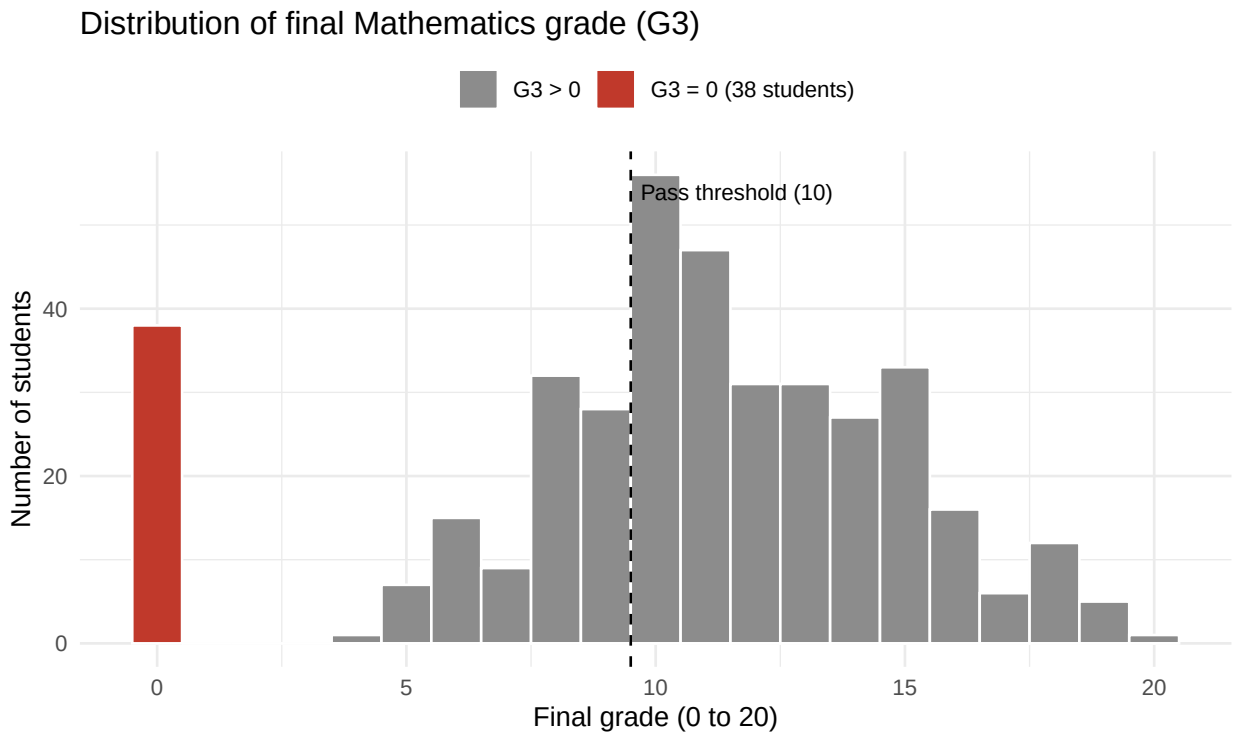
```r
table(students$failures)
```

```
##
##   0   1   2   3
## 312  50  17  16
```

```r
table(students$schoolsup)
```

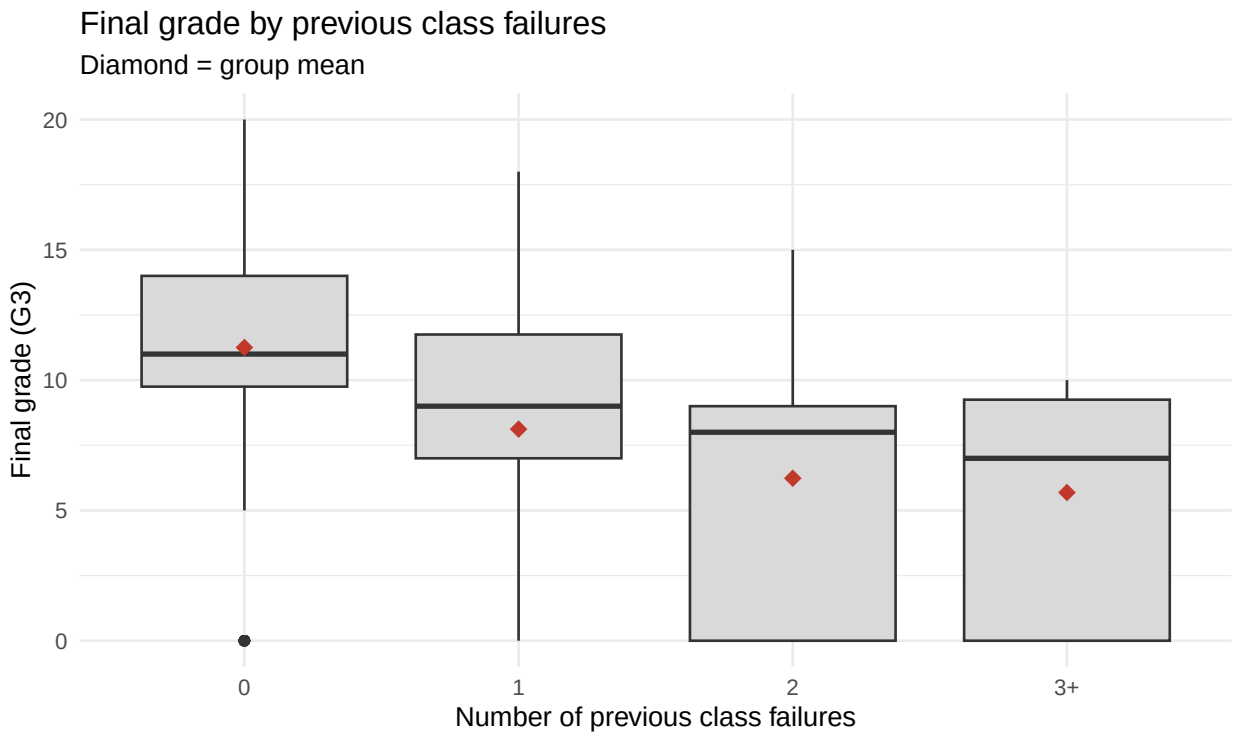
```
##
##  no yes
## 344  51
```

```r
# Figure 1: distribution of the final grade, zero group highlighted.
ggplot(students, aes(x = G3, fill = G3 == 0)) +
  geom_histogram(binwidth = 1, boundary = -0.5, colour = "white") +
  geom_vline(xintercept = 9.5, linetype = 2) +
  annotate("text", x = 9.7, y = 54, label = "Pass threshold (10)",
           hjust = 0, size = 3.3) +
  scale_fill_manual(values = c(`FALSE` = "grey55", `TRUE` = "#C0392B"),
                    labels = c("G3 > 0", "G3 = 0 (38 students)"),
                    name = NULL) +
  labs(title = "Distribution of final Mathematics grade (G3)",
       x = "Final grade (0 to 20)", y = "Number of students") +
  theme(legend.position = "top")
```



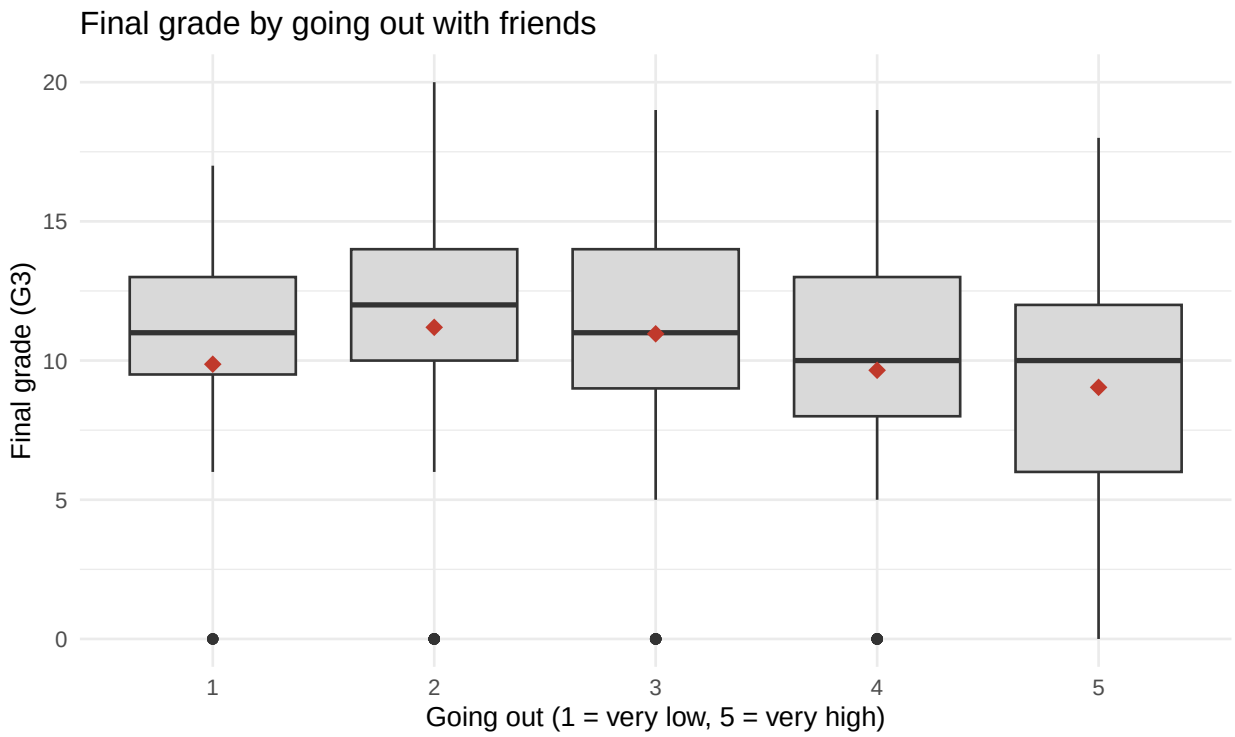

```
save_fig("fig1_g3_distribution")

# Figure 2: final grade by number of previous class failures.
ggplot(students, aes(x = factor(failures), y = G3)) +
  geom_boxplot(fill = "grey85") +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3,
               colour = "#C0392B") +
  scale_x_discrete(labels = c("0", "1", "2", "3+")) +
  labs(title = "Final grade by previous class failures",
       subtitle = "Diamond = group mean",
       x = "Number of previous class failures", y = "Final grade (G3)")
```



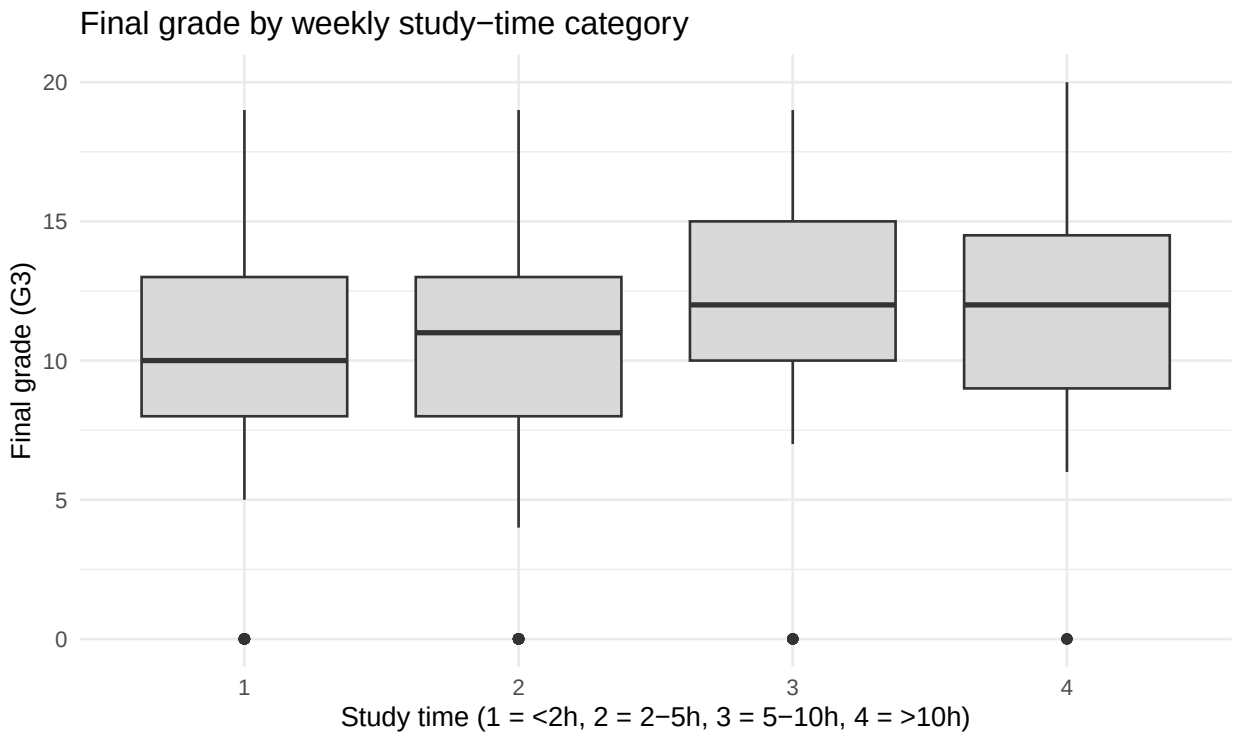

```
save_fig("fig2_g3_by_failures", width = 5, height = 4)

# Figure 3: final grade by going out with friends.
ggplot(students, aes(x = factor(goout), y = G3)) +
  geom_boxplot(fill = "grey85") +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3,
               colour = "#C0392B") +
  labs(title = "Final grade by going out with friends",
       x = "Going out (1 = very low, 5 = very high)", y = "Final grade (G3)")
```



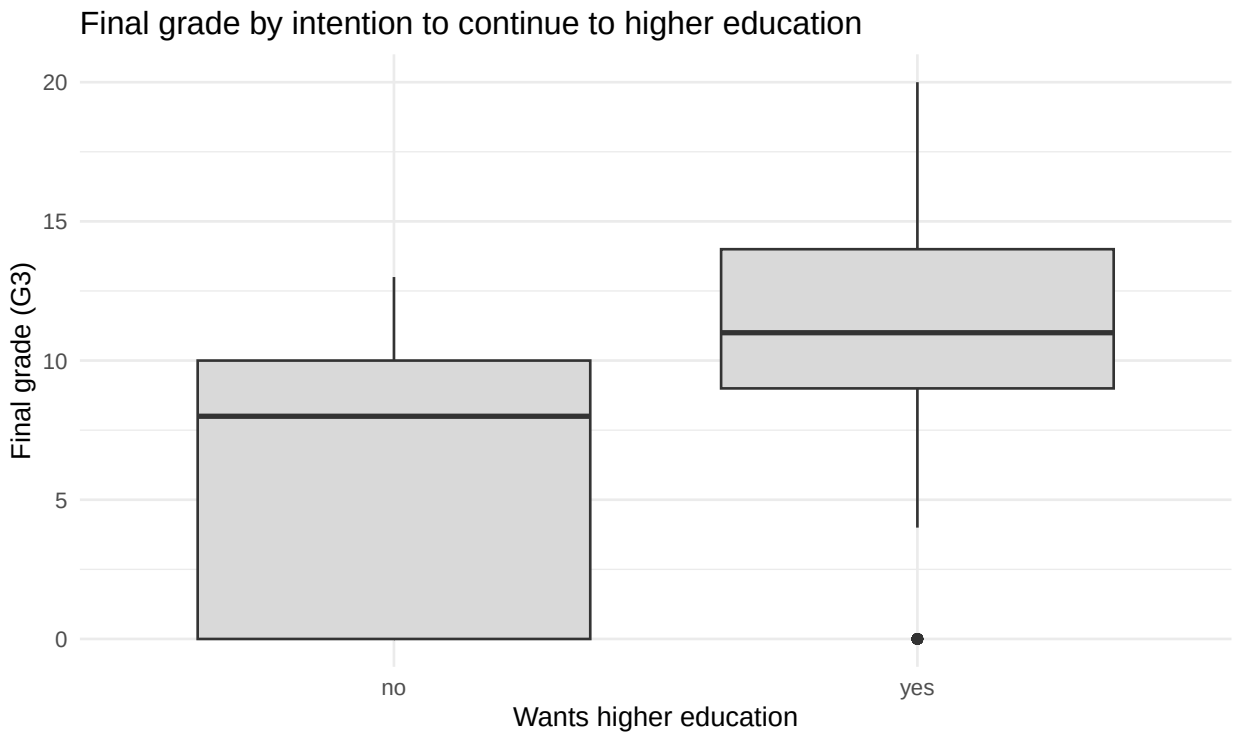

```
save_fig("fig3_g3_by_goout", width = 5, height = 4)

# Additional exploratory plots (from the starter script).
ggplot(students, aes(x = factor(studytime), y = G3)) +
  geom_boxplot(fill = "grey85") +
  labs(title = "Final grade by weekly study-time category",
       x = "Study time (1 = <2h, 2 = 2-5h, 3 = 5-10h, 4 = >10h)",
       y = "Final grade (G3)")
```




```
save_fig("figA_g3_by_studytime", width = 5, height = 4)

ggplot(students, aes(x = higher, y = G3)) +
  geom_boxplot(fill = "grey85") +
  labs(title = "Final grade by intention to continue to higher education",
       x = "Wants higher education", y = "Final grade (G3)")
```




```
save_fig("figA_g3_by_higher", width = 5, height = 4)

pass_by_sex <- students %>%
  count(sex, pass) %>%
  group_by(sex) %>%
  mutate(percent = 100 * n / sum(n))
print(pass_by_sex)
```

```
## # A tibble: 4 x 4
## # Groups:   sex [2]
##   sex   pass      n percent
##   <fct> <fct> <int>   <dbl>
## 1 F     Fail     75    36.1
## 2 F     Pass    133    63.9
## 3 M     Fail     55    29.4
## 4 M     Pass    132    70.6
```

Question 3. Investigate relationships

```
cor_vars <- students %>%
  select(age, Medu, Fedu, traveltime, studytime, failures,
         famrel, freetime, goout, Dalc, Walc, health, absences, G3)

# Pearson treats codes as interval measurements; Spearman only uses ranks and
# is therefore more defensible for ordered categories. We report both.
cor_pearson  <- cor(cor_vars, method = "pearson")
cor_spearman <- cor(cor_vars, method = "spearman")
```

```r
g3_cor <- data.frame(
  variable = rownames(cor_pearson),
  pearson  = cor_pearson[, "G3"],
  spearman = cor_spearman[, "G3"]
) %>%
  filter(variable != "G3") %>%
  arrange(pearson)
print(g3_cor, digits = 2)
```

```
##              variable pearson spearman
## failures     failures  -0.360   -0.361
## age               age  -0.162   -0.173
## goout           goout  -0.133   -0.166
## traveltime traveltime  -0.117   -0.121
## health         health  -0.061   -0.048
## Dalc             Dalc  -0.055   -0.121
## Walc             Walc  -0.052   -0.104
## freetime     freetime   0.011   -0.005
## absences     absences   0.034    0.018
## famrel         famrel   0.051    0.055
## studytime   studytime   0.098    0.105
## Fedu             Fedu   0.152    0.170
## Medu             Medu   0.217    0.225
```

```r
write.csv(g3_cor, "results/correlations_with_G3.csv", row.names = FALSE)

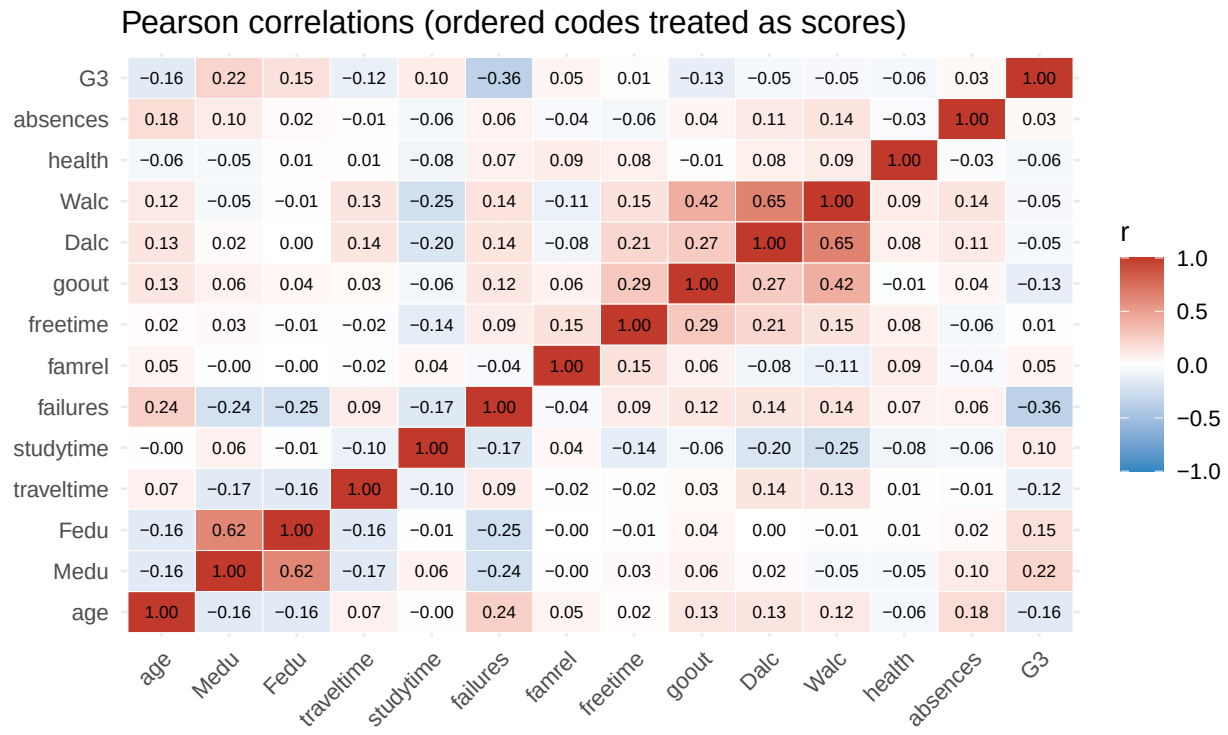
# Notable correlations between predictors (|r| >= 0.4).
round(cor_pearson, 2)
```

```
##              age  Medu  Fedu traveltime studytime failures famrel freetime
## age         1.00 -0.16 -0.16       0.07      0.00     0.24   0.05     0.02
## Medu       -0.16  1.00  0.62      -0.17      0.06    -0.24   0.00     0.03
## Fedu       -0.16  0.62  1.00      -0.16     -0.01    -0.25   0.00    -0.01
## traveltime  0.07 -0.17 -0.16       1.00     -0.10     0.09  -0.02    -0.02
## studytime   0.00  0.06 -0.01      -0.10      1.00    -0.17   0.04    -0.14
## failures    0.24 -0.24 -0.25       0.09     -0.17     1.00  -0.04     0.09
## famrel      0.05  0.00  0.00      -0.02      0.04    -0.04   1.00     0.15
## freetime    0.02  0.03 -0.01      -0.02     -0.14     0.09   0.15     1.00
## goout       0.13  0.06  0.04       0.03     -0.06     0.12   0.06     0.29
## Dalc        0.13  0.02  0.00       0.14     -0.20     0.14  -0.08     0.21
## Walc        0.12 -0.05 -0.01       0.13     -0.25     0.14  -0.11     0.15
## health     -0.06 -0.05  0.01       0.01     -0.08     0.07   0.09     0.08
## absences    0.18  0.10  0.02      -0.01     -0.06     0.06  -0.04    -0.06
## G3         -0.16  0.22  0.15      -0.12      0.10    -0.36   0.05     0.01
##             goout  Dalc  Walc health absences    G3
## age          0.13  0.13  0.12  -0.06     0.18 -0.16
## Medu         0.06  0.02 -0.05  -0.05     0.10  0.22
## Fedu         0.04  0.00 -0.01   0.01     0.02  0.15
## traveltime   0.03  0.14  0.13   0.01    -0.01 -0.12
## studytime   -0.06 -0.20 -0.25  -0.08    -0.06  0.10
## failures     0.12  0.14  0.14   0.07     0.06 -0.36
## famrel       0.06 -0.08 -0.11   0.09    -0.04  0.05
```

```
## freetime    0.29  0.21  0.15   0.08    -0.06  0.01
## goout       1.00  0.27  0.42  -0.01     0.04 -0.13
## Dalc        0.27  1.00  0.65   0.08     0.11 -0.05
## Walc        0.42  0.65  1.00   0.09     0.14 -0.05
## health     -0.01  0.08  0.09   1.00    -0.03 -0.06
## absences    0.04  0.11  0.14  -0.03     1.00  0.03
## G3         -0.13 -0.05 -0.05  -0.06     0.03  1.00
```

```r
# Medu-Fedu (about 0.62) and Dalc-Walc (about 0.65) overlap strongly, so only
# one of each pair is a candidate for the model.

# Figure 4: correlation heatmap (ggplot version, saved for the report).
cor_long <- as.data.frame(as.table(cor_pearson))
names(cor_long) <- c("var1", "var2", "r")
ggplot(cor_long, aes(var1, var2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 2.3) +
  scale_fill_gradient2(low = "#2E86C1", mid = "white", high = "#C0392B",
                       limits = c(-1, 1), name = "r") +
  labs(title = "Pearson correlations (ordered codes treated as scores)",
       x = NULL, y = NULL) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
save_fig("fig4_correlation_heatmap", width = 7, height = 6)
```

Contingency tables with conditional (column) percentages

```
pass_table <- function(var) {
  tab <- table(students$pass, students[[var]])
  pct <- round(prop.table(tab, margin = 2) * 100, 1)
  print(tab)
  print(pct)
  invisible(pct)
}

pass_table("higher")      # wants higher education: yes vs no
```

```
##
##         no yes
##   Fail  13 117
##   Pass   7 258
##
##          no  yes
##   Fail 65.0 31.2
##   Pass 35.0 68.8
```

```
pass_table("schoolsup")   # extra educational support
```

```
##
##         no yes
##   Fail 107  23
##   Pass 237  28
##
##          no  yes
##   Fail 31.1 45.1
##   Pass 68.9 54.9
```

```
pass_table("sex")
```

```
##
##          F   M
##   Fail  75  55
##   Pass 133 132
##
##           F    M
##   Fail 36.1 29.4
##   Pass 63.9 70.6
```

```
pass_table("romantic")
```

```
##
##         no yes
##   Fail  78  52
##   Pass 185  80
##
```

```
##          no  yes
##   Fail 29.7 39.4
##   Pass 70.3 60.6
```

```
# Previous failures grouped as any vs none.
students$any_failure <- factor(ifelse(students$failures > 0, "1 or more", "None"),
                               levels = c("None", "1 or more"))
tab_fail <- table(students$pass, students$any_failure)
print(tab_fail)
```

```
##
##        None 1 or more
##   Fail   78        52
##   Pass  234        31
```

```
round(prop.table(tab_fail, margin = 2) * 100, 1)
```

```
##
##        None 1 or more
##   Fail 25.0      62.7
##   Pass 75.0      37.3
```

```
# Chi-squared tests as a rough check of association (expected counts for
# 'higher = no' are small, so Fisher's exact test is used there).
chisq.test(tab_fail)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab_fail
## X-squared = 40, df = 1, p-value = 2e-10
```

```
fisher.test(table(students$pass, students$higher))
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table(students$pass, students$higher)
## p-value = 0.003
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##   1.468 12.405
## sample estimates:
## odds ratio
##      4.079
```

```
chisq.test(table(students$pass, students$schoolsup))
```

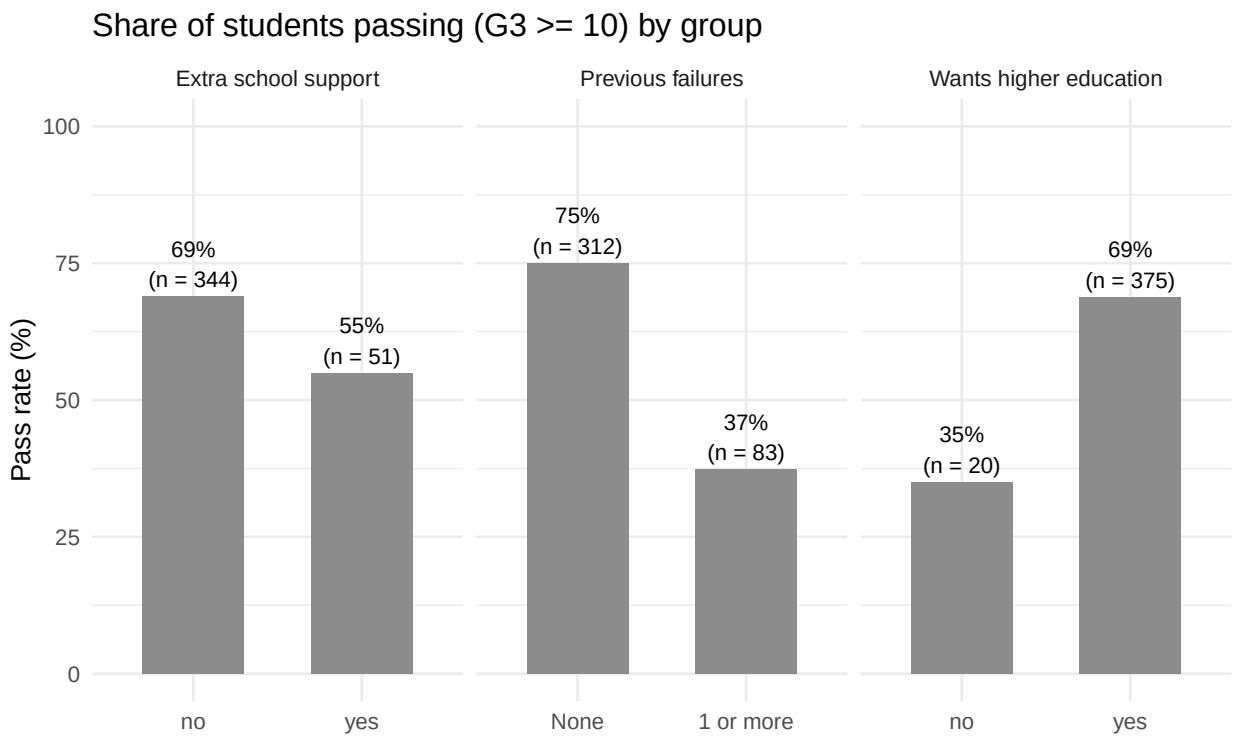
```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(students$pass, students$schoolsup)
## X-squared = 3.3, df = 1, p-value = 0.07
```

```r
# Figure 5: pass rate by key groups, for the report and presentation.
pass_rates <- bind_rows(
  students %>% group_by(level = any_failure) %>%
    summarise(rate = mean(pass == "Pass"), n = n()) %>%
    mutate(characteristic = "Previous failures"),
  students %>% group_by(level = higher) %>%
    summarise(rate = mean(pass == "Pass"), n = n()) %>%
    mutate(characteristic = "Wants higher education", level = as.character(level)),
  students %>% group_by(level = schoolsup) %>%
    summarise(rate = mean(pass == "Pass"), n = n()) %>%
    mutate(characteristic = "Extra school support", level = as.character(level))
) %>%
  mutate(level = factor(as.character(level),
                        levels = c("None", "1 or more", "no", "yes")))
print(pass_rates)
```

```
## # A tibble: 6 x 4
##   level      rate     n characteristic
##   <fct>     <dbl> <int> <chr>
## 1 None      0.75    312 Previous failures
## 2 1 or more 0.373    83 Previous failures
## 3 no        0.35     20 Wants higher education
## 4 yes       0.688   375 Wants higher education
## 5 no        0.689   344 Extra school support
## 6 yes       0.549    51 Extra school support
```

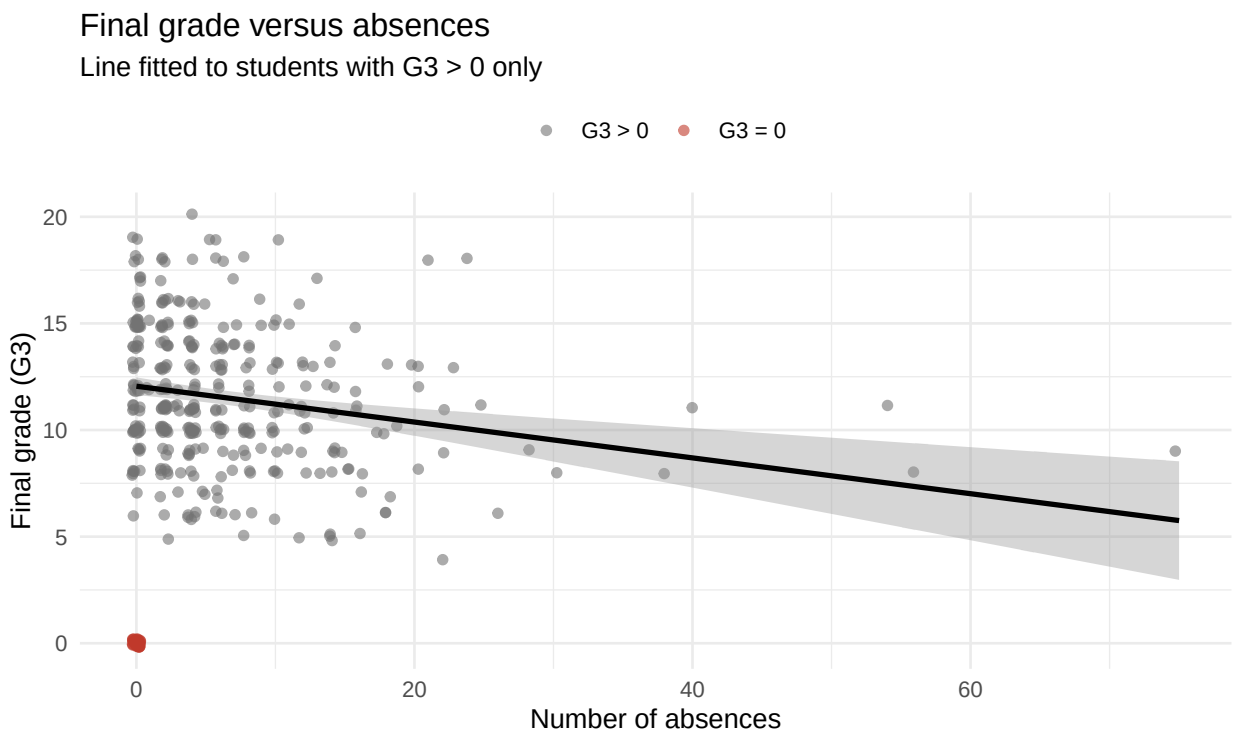
```r
write.csv(pass_rates, "results/pass_rates_by_group.csv", row.names = FALSE)

ggplot(pass_rates, aes(x = level, y = rate * 100)) +
  geom_col(fill = "grey55", width = 0.6) +
  geom_text(aes(label = paste0(round(rate * 100), "%\n(n = ", n, ")")),
            vjust = -0.2, size = 3) +
  facet_wrap(~ characteristic, scales = "free_x") +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = "Share of students passing (G3 >= 10) by group",
       x = NULL, y = "Pass rate (%)")
```




```
save_fig("fig5_pass_rates", width = 7, height = 3.6)

# Absences versus final grade, coloured by the zero group: shows why the
# overall correlation with absences is close to zero.
ggplot(students, aes(x = absences, y = G3, colour = G3 == 0)) +
  geom_jitter(alpha = 0.6, width = 0.3, height = 0.2) +
  geom_smooth(data = students[!zero_g3, ], aes(group = 1), method = "lm",
              se = TRUE, colour = "black") +
  scale_colour_manual(values = c(`FALSE` = "grey45", `TRUE` = "#C0392B"),
                      labels = c("G3 > 0", "G3 = 0"), name = NULL) +
  labs(title = "Final grade versus absences",
       subtitle = "Line fitted to students with G3 > 0 only",
       x = "Number of absences", y = "Final grade (G3)") +
  theme(legend.position = "top")
```




```
save_fig("fig6_absences_vs_g3", width = 6, height = 4)

# Choice of predictors for modelling (justified in the report):
#   failures  : strongest correlation with G3 (about -0.36)
#   Medu      : strongest positive correlation (about 0.22); chosen over Fedu
#   goout     : negative association (about -0.13); chosen over Dalc/Walc
#   studytime : expected to matter; weak but positive
#   absences  : behaviour of interest; distorted by the zero group
#   sex       : pass rates differ by sex
#   higher    : very large pass-rate gap, but only 20 students answer 'no'
# schoolsup is NOT included in the main model: extra support is given to
# weaker students, so its association with grades reflects selection, not
# the effect of support (reverse causation).
```

Question 4. Build and interpret statistical models

```
# Baseline: the starter-script model, kept for comparison.
baseline_model <- lm(G3 ~ studytime + failures + absences + sex + higher,
                     data = students)
summary(baseline_model)
```

```
##
## Call:
## lm(formula = G3 ~ studytime + failures + absences + sex + higher,
##     data = students)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -12.563  -1.977   0.286   2.926   9.173
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.3212     1.2288    5.96  5.7e-09 ***
## studytime     0.4246     0.2716    1.56   0.1189
## failures     -2.0290     0.3025   -6.71  7.0e-11 ***
## absences      0.0438     0.0268    1.64   0.1027
## sexM          1.4840     0.4512    3.29   0.0011 **
## higheryes     2.0597     1.0306    2.00   0.0464 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.22 on 389 degrees of freedom
## Multiple R-squared:  0.164,  Adjusted R-squared:  0.153
## F-statistic: 15.2 on 5 and 389 DF,  p-value: 1.13e-13
```

Main multiple linear regression

```r
# Reference categories: sex = F (female), higher = no.
linear_model <- lm(
  G3 ~ failures + Medu + goout + studytime + absences + sex + higher,
  data = students
)
summary(linear_model)
```

```
##
## Call:
## lm(formula = G3 ~ failures + Medu + goout + studytime + absences +
##     sex + higher, data = students)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -13.494  -2.061   0.489   2.729   8.856
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.7446     1.3789    5.62  3.7e-08 ***
## failures     -1.8004     0.3078   -5.85  1.0e-08 ***
## Medu          0.4922     0.2027    2.43   0.0156 *
## goout        -0.4557     0.1916   -2.38   0.0179 *
## studytime     0.3750     0.2692    1.39   0.1644
## absences      0.0374     0.0267    1.40   0.1624
## sexM          1.4109     0.4507    3.13   0.0019 **
## higheryes     1.7811     1.0271    1.73   0.0837 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.17 on 387 degrees of freedom
## Multiple R-squared:  0.186,  Adjusted R-squared:  0.171
## F-statistic: 12.6 on 7 and 387 DF,  p-value: 1.28e-14
```

```r
linear_coef <- tidy(linear_model, conf.int = TRUE)
print(linear_coef)
```

```
## # A tibble: 8 x 7
##   term        estimate std.error statistic      p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>        <dbl>    <dbl>     <dbl>
## 1 (Intercept)   7.74      1.38        5.62 0.0000000373   5.03     10.5
## 2 failures     -1.80      0.308      -5.85 0.0000000105  -2.41     -1.20
## 3 Medu          0.492     0.203       2.43 0.0156         0.0936    0.891
## 4 goout        -0.456     0.192      -2.38 0.0179        -0.832    -0.0789
## 5 studytime     0.375     0.269       1.39 0.164         -0.154     0.904
## 6 absences      0.0374    0.0267      1.40 0.162         -0.0151    0.0900
## 7 sexM          1.41      0.451       3.13 0.00188        0.525     2.30
## 8 higheryes     1.78      1.03        1.73 0.0837        -0.238     3.80
```

```r
write.csv(linear_coef, "results/linear_model_coefficients.csv", row.names = FALSE)

model_fit <- bind_rows(
  glance(baseline_model) %>% mutate(model = "Baseline (starter)"),
  glance(linear_model)   %>% mutate(model = "Main model")
) %>% select(model, r.squared, adj.r.squared, sigma, AIC, nobs)
print(model_fit)
```

```
## # A tibble: 2 x 6
##   model              r.squared adj.r.squared sigma   AIC  nobs
##   <chr>                  <dbl>         <dbl> <dbl> <dbl> <int>
## 1 Baseline (starter)     0.164         0.153  4.22 2266.   395
## 2 Main model             0.186         0.171  4.17 2259.   395
```

```r
write.csv(model_fit, "results/linear_model_fit.csv", row.names = FALSE)

# Illustration: predicted G3 for a 'typical' student with 0 vs 2 failures,
# other predictors at their median / most common level.
typical <- data.frame(failures = c(0, 2), Medu = 3, goout = 3, studytime = 2,
                      absences = 4, sex = "F", higher = "yes")
predict(linear_model, newdata = typical, interval = "confidence")
```

```
##      fit   lwr    upr
## 1 10.535 9.895 11.175
## 2  6.934 5.709  8.159
```

Logistic regression for pass/fail

```r
# glm() models the probability of the second level, Pass.
logistic_model <- glm(
  pass ~ failures + Medu + goout + studytime + absences + sex + higher,
  data = students, family = binomial
)
summary(logistic_model)
```

```
##
## Call:
## glm(formula = pass ~ failures + Medu + goout + studytime + absences +
##     sex + higher, family = binomial, data = students)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.8052     0.7534    1.07   0.2852
## failures     -0.8677     0.1805   -4.81  1.5e-06 ***
## Medu          0.1015     0.1128    0.90   0.3682
## goout        -0.3495     0.1083   -3.23   0.0012 **
## studytime     0.0894     0.1531    0.58   0.5591
## absences     -0.0159     0.0137   -1.17   0.2440
## sexM          0.5547     0.2556    2.17   0.0300 *
## higheryes     0.7568     0.5605    1.35   0.1770
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 500.50  on 394  degrees of freedom
## Residual deviance: 438.47  on 387  degrees of freedom
## AIC: 454.5
##
## Number of Fisher Scoring iterations: 4
```

```
logistic_coef <- tidy(logistic_model, conf.int = TRUE)
logistic_or   <- tidy(logistic_model, conf.int = TRUE, exponentiate = TRUE)
print(logistic_or)
```

```
## # A tibble: 8 x 7
##   term        estimate std.error statistic    p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>      <dbl>    <dbl>     <dbl>
## 1 (Intercept)    2.24     0.753      1.07  0.285         0.508     9.90
## 2 failures       0.420    0.181     -4.81  0.00000153    0.290     0.590
## 3 Medu           1.11     0.113      0.900 0.368         0.887     1.38
## 4 goout          0.705    0.108     -3.23  0.00125       0.568     0.870
## 5 studytime      1.09     0.153      0.584 0.559         0.813     1.48
## 6 absences       0.984    0.0137    -1.17  0.244         0.958     1.01
## 7 sexM           1.74     0.256      2.17  0.0300        1.06      2.89
## 8 higheryes      2.13     0.561      1.35  0.177         0.715     6.62
```

```
write.csv(logistic_or, "results/logistic_odds_ratios.csv", row.names = FALSE)

# Predicted probabilities of passing for the same two 'typical' students.
# This translates odds ratios into probabilities for the presentation.
typical$p_pass <- predict(logistic_model, newdata = typical, type = "response")
print(typical)
```

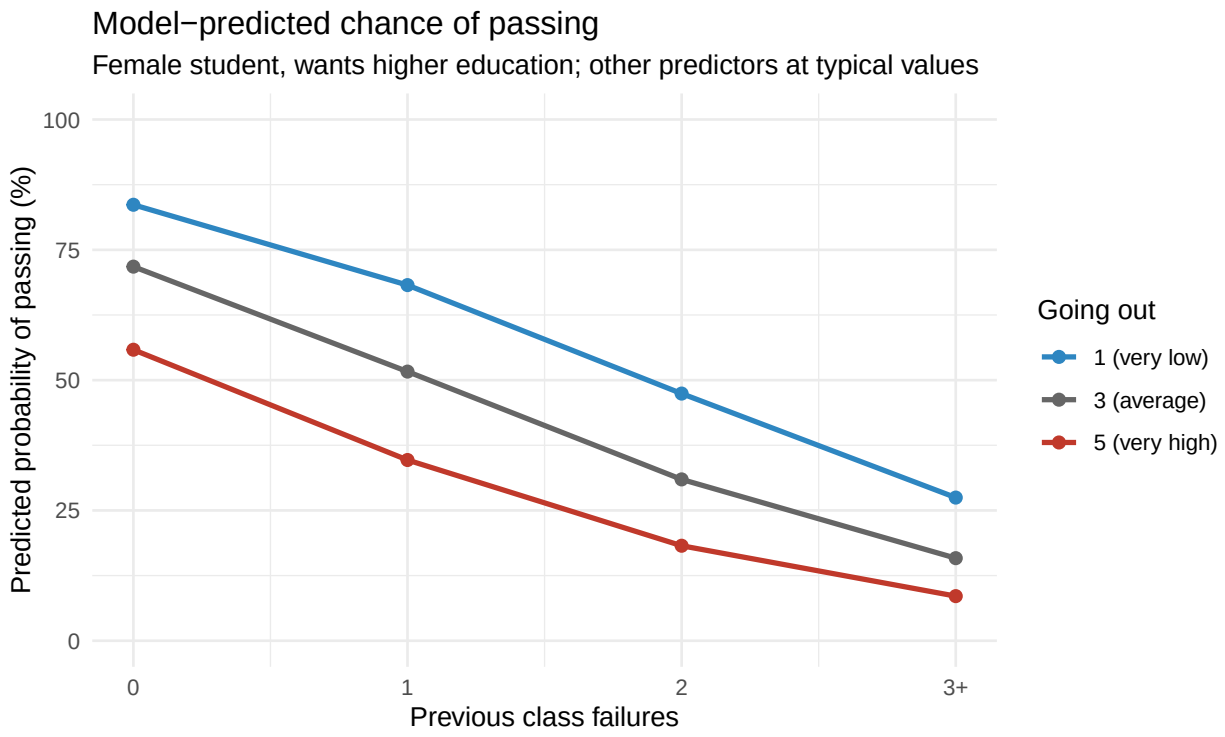
```
##   failures Medu goout studytime absences sex higher p_pass
## 1        0    3     3         2        4   F    yes 0.7177
## 2        2    3     3         2        4   F    yes 0.3095
```

```r
# Predicted probability of passing across failures and going out.
prob_grid <- expand.grid(failures = 0:3, goout = c(1, 3, 5), Medu = 3,
                         studytime = 2, absences = 4, sex = "F", higher = "yes")
prob_grid$p_pass <- predict(logistic_model, newdata = prob_grid, type = "response")
print(prob_grid)
```

```
##    failures goout Medu studytime absences sex higher  p_pass
## 1         0     1    3         2        4   F    yes 0.83646
## 2         1     1    3         2        4   F    yes 0.68231
## 3         2     1    3         2        4   F    yes 0.47419
## 4         3     1    3         2        4   F    yes 0.27466
## 5         0     3    3         2        4   F    yes 0.71773
## 6         1     3    3         2        4   F    yes 0.51637
## 7         2     3    3         2        4   F    yes 0.30954
## 8         3     3    3         2        4   F    yes 0.15843
## 9         0     5    3         2        4   F    yes 0.55831
## 10        1     5    3         2        4   F    yes 0.34674
## 11        2     5    3         2        4   F    yes 0.18225
## 12        3     5    3         2        4   F    yes 0.08558
```

```r
write.csv(prob_grid, "results/predicted_pass_probabilities.csv", row.names = FALSE)

# Figure 7: predicted pass probability by failures (for presentation).
ggplot(prob_grid, aes(x = failures, y = p_pass * 100,
                      colour = factor(goout), group = factor(goout))) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_colour_manual(values = c("1" = "#2E86C1", "3" = "grey40", "5" = "#C0392B"),
                      labels = c("1 (very low)", "3 (average)", "5 (very high)"),
                      name = "Going out") +
  scale_x_continuous(breaks = 0:3, labels = c("0", "1", "2", "3+")) +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = "Model-predicted chance of passing",
       subtitle = "Female student, wants higher education; other predictors at typical values",
       x = "Previous class failures", y = "Predicted probability of passing (%)")
```




```
save_fig("fig7_predicted_pass_probability", width = 6, height = 4)

# Simple classification check at a 0.5 cut-off. Only 'Fail' predictions are
# useful for an early-warning reading, so sensitivity for Fail is reported.
students$p_pass <- predict(logistic_model, type = "response")
students$pred_class <- factor(ifelse(students$p_pass >= 0.5, "Pass", "Fail"),
                              levels = c("Fail", "Pass"))
conf_mat <- table(Observed = students$pass, Predicted = students$pred_class)
print(conf_mat)
```

```
##         Predicted
## Observed Fail Pass
##     Fail   45   85
##     Pass   22  243
```

```
accuracy <- sum(diag(conf_mat)) / sum(conf_mat)
fail_detected <- conf_mat["Fail", "Fail"] / sum(conf_mat["Fail", ])
c(accuracy = accuracy, share_of_fails_detected = fail_detected,
  no_information_rate = max(prop.table(table(students$pass))))
```

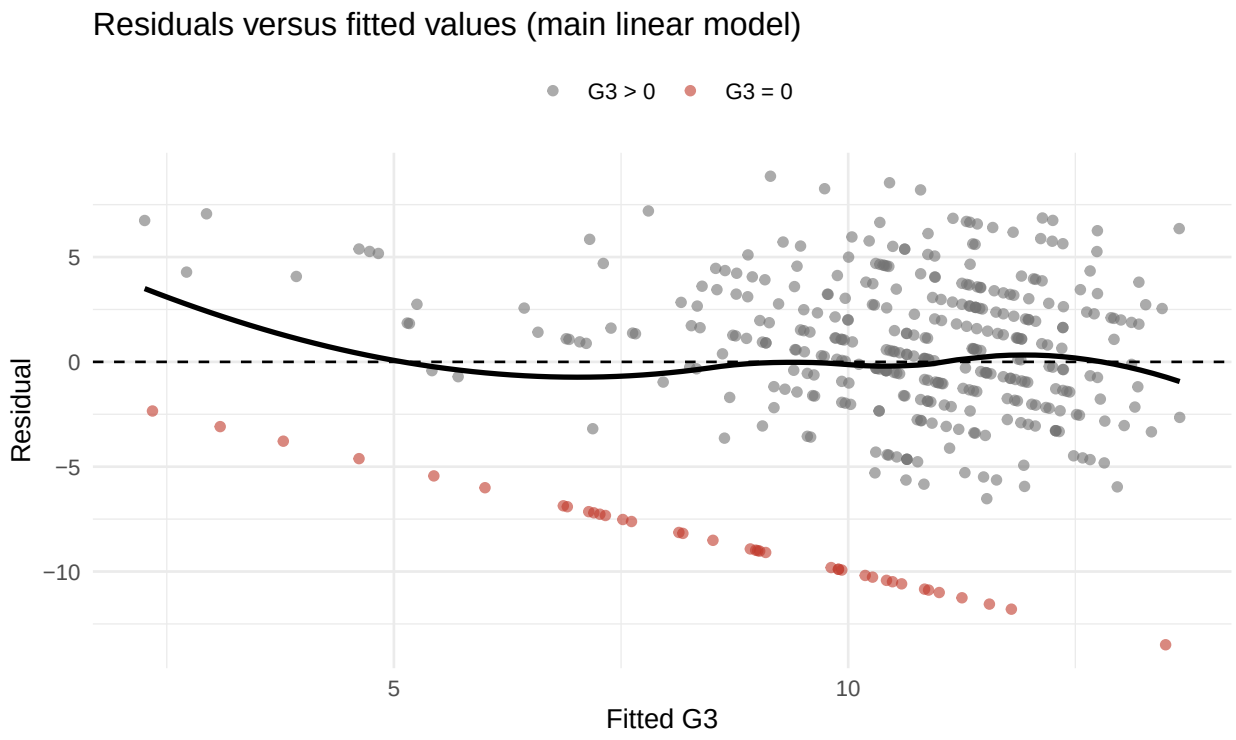
```
##               accuracy share_of_fails_detected     no_information_rate
##                 0.7291                 0.3462                  0.6709
```

Question 5. Evaluate the analysis

Residuals versus fitted

```
diagnostic_data <- augment(linear_model) %>%
  mutate(zero_grade = G3 == 0)

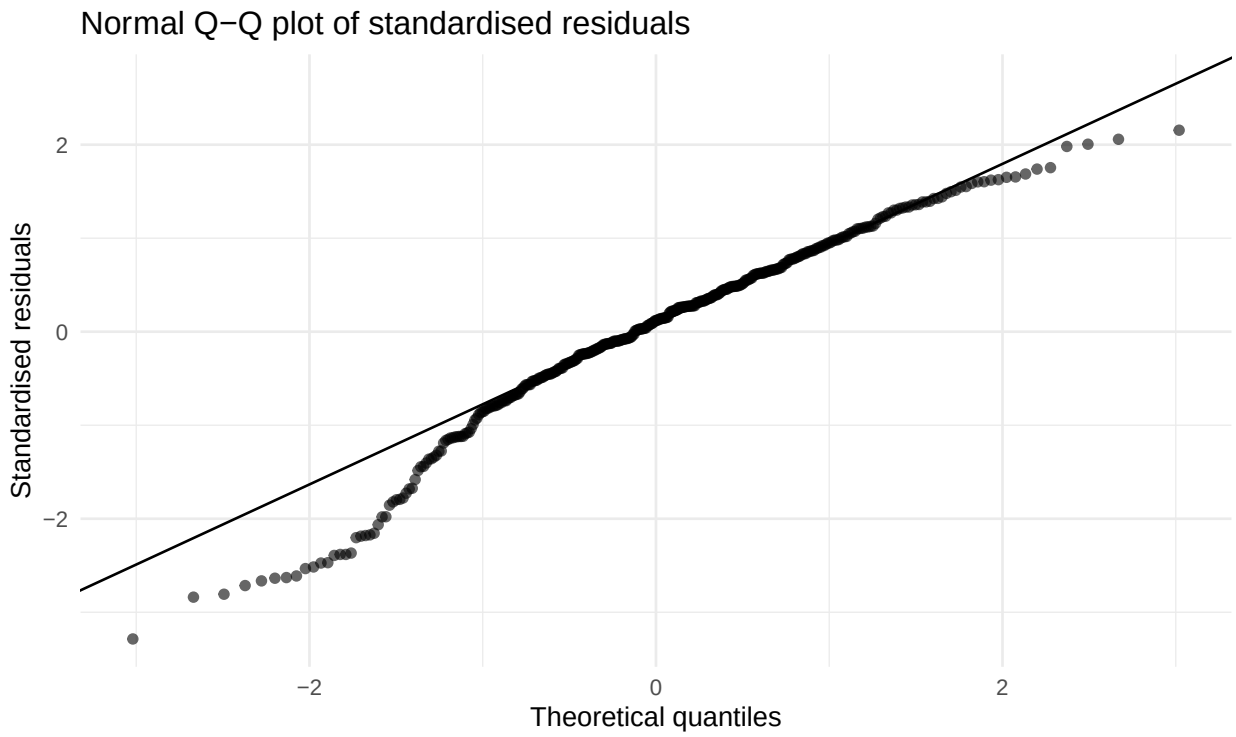
ggplot(diagnostic_data, aes(x = .fitted, y = .resid, colour = zero_grade)) +
  geom_point(alpha = 0.6) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_smooth(aes(group = 1), se = FALSE, colour = "black", method = "loess") +
  scale_colour_manual(values = c(`FALSE` = "grey45", `TRUE` = "#C0392B"),
                      labels = c("G3 > 0", "G3 = 0"), name = NULL) +
  labs(title = "Residuals versus fitted values (main linear model)",
       x = "Fitted G3", y = "Residual") +
  theme(legend.position = "top")
```




```
save_fig("fig8_residuals_vs_fitted", width = 6, height = 4)

# The G3 = 0 students form a diagonal band of large negative residuals
# (residual = -fitted). This is a clear violation of the linear-model
# assumptions for that group.

# Normal Q-Q plot of residuals (supplementary).
ggplot(diagnostic_data, aes(sample = .std.resid)) +
  stat_qq(alpha = 0.6) + stat_qq_line() +
  labs(title = "Normal Q-Q plot of standardised residuals",
       x = "Theoretical quantiles", y = "Standardised residuals")
```




```
save_fig("figA_qq_plot", width = 5, height = 4)
```

Overlap between predictors

```
manual_vif <- function(model) {
  X <- model.matrix(model)[, -1, drop = FALSE]
  vif_values <- sapply(seq_len(ncol(X)), function(i) {
    r2 <- summary(lm(X[, i] ~ X[, -i, drop = FALSE]))$r.squared
    1 / (1 - r2)
  })
  setNames(vif_values, colnames(X))
}
vif_values <- manual_vif(linear_model)
round(vif_values, 2)   # all close to 1: little overlap among predictors
```

```
##  failures      Medu     goout studytime  absences      sexM higheryes
##      1.19      1.12      1.03      1.16      1.04      1.15      1.15
```

```
round(cor(students %>% select(failures, Medu, goout, studytime, absences)), 2)
```

```
##           failures  Medu goout studytime absences
## failures      1.00 -0.24  0.12     -0.17     0.06
## Medu         -0.24  1.00  0.06      0.06     0.10
## goout         0.12  0.06  1.00     -0.06     0.04
## studytime    -0.17  0.06 -0.06      1.00    -0.06
## absences      0.06  0.10  0.04     -0.06     1.00
```

Uncertainty

```
print(linear_coef)
```

```
## # A tibble: 8 x 7
##   term        estimate std.error statistic      p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>        <dbl>    <dbl>     <dbl>
## 1 (Intercept)   7.74      1.38        5.62 0.0000000373   5.03     10.5
## 2 failures     -1.80      0.308      -5.85 0.0000000105  -2.41     -1.20
## 3 Medu          0.492     0.203       2.43 0.0156         0.0936    0.891
## 4 goout        -0.456     0.192      -2.38 0.0179        -0.832    -0.0789
## 5 studytime     0.375     0.269       1.39 0.164         -0.154     0.904
## 6 absences      0.0374    0.0267      1.40 0.162         -0.0151    0.0900
## 7 sexM          1.41      0.451       3.13 0.00188        0.525     2.30
## 8 higheryes     1.78      1.03        1.73 0.0837        -0.238     3.80
```

```
print(logistic_or)
```

```
## # A tibble: 8 x 7
##   term        estimate std.error statistic    p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>      <dbl>    <dbl>     <dbl>
## 1 (Intercept)    2.24      0.753      1.07  0.285         0.508     9.90
## 2 failures       0.420     0.181     -4.81  0.00000153    0.290     0.590
## 3 Medu           1.11      0.113      0.900 0.368         0.887     1.38
## 4 goout          0.705     0.108     -3.23  0.00125       0.568     0.870
## 5 studytime      1.09      0.153      0.584 0.559         0.813     1.48
## 6 absences       0.984     0.0137    -1.17  0.244         0.958     1.01
## 7 sexM           1.74      0.256      2.17  0.0300        1.06      2.89
## 8 higheryes      2.13      0.561      1.35  0.177         0.715     6.62
```

Sensitivity analysis: exclude the 38 students with G3 = 0

```
students_nz <- students %>% filter(G3 > 0)

linear_model_nz <- lm(
  G3 ~ failures + Medu + goout + studytime + absences + sex + higher,
  data = students_nz
)
summary(linear_model_nz)
```

```
##
## Call:
## lm(formula = G3 ~ failures + Medu + goout + studytime + absences +
##     sex + higher, data = students_nz)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.509 -1.980 -0.273  2.104  9.028
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.0098     1.0867   10.13  < 2e-16 ***
## failures     -0.9758     0.2524   -3.87  0.00013 ***
## Medu          0.4642     0.1505    3.08  0.00220 **
## goout        -0.4621     0.1458   -3.17  0.00166 **
## studytime     0.4162     0.2000    2.08  0.03814 *
## absences     -0.0658     0.0197   -3.33  0.00095 ***
## sexM          0.8106     0.3328    2.44  0.01537 *
## higheryes     0.0917     0.8618    0.11  0.91536
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.95 on 349 degrees of freedom
## Multiple R-squared:  0.18,   Adjusted R-squared:  0.164
## F-statistic: 10.9 on 7 and 349 DF,  p-value: 1.68e-12
```

```
logistic_model_nz <- glm(
  pass ~ failures + Medu + goout + studytime + absences + sex + higher,
  data = students_nz, family = binomial
)

sensitivity <- bind_rows(
  tidy(linear_model, conf.int = TRUE)    %>% mutate(model = "Linear, all 395"),
  tidy(linear_model_nz, conf.int = TRUE) %>% mutate(model = "Linear, G3 > 0 (357)"),
  tidy(logistic_model, conf.int = TRUE, exponentiate = TRUE) %>%
    mutate(model = "Logistic OR, all 395"),
  tidy(logistic_model_nz, conf.int = TRUE, exponentiate = TRUE) %>%
    mutate(model = "Logistic OR, G3 > 0 (357)")
) %>% filter(term != "(Intercept)")
print(sensitivity, n = 40)
```

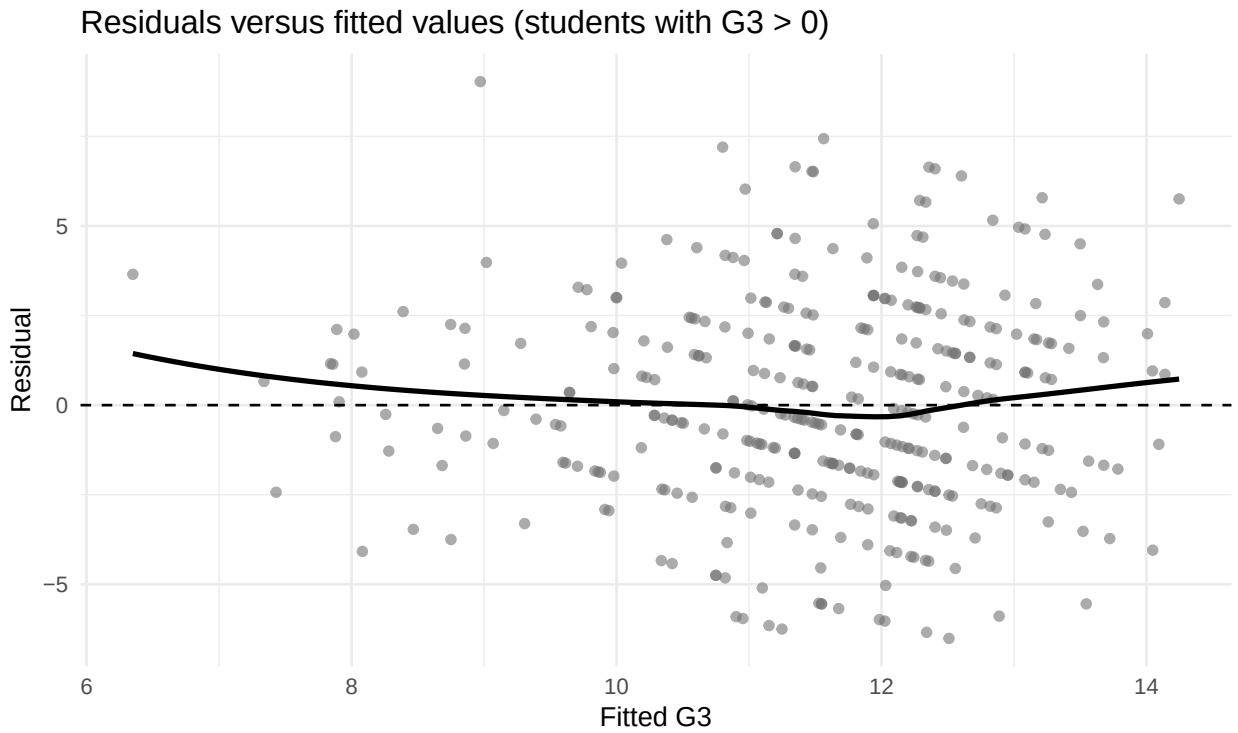
```
## # A tibble: 28 x 8
##    term      estimate std.error statistic      p.value conf.low conf.high model
##    <chr>        <dbl>     <dbl>     <dbl>        <dbl>    <dbl>     <dbl> <chr>
##  1 failures   -1.80      0.308     -5.85  0.0000000105  -2.41     -1.20   Linea~
##  2 Medu        0.492     0.203      2.43  0.0156         0.0936    0.891  Linea~
##  3 goout      -0.456     0.192     -2.38  0.0179        -0.832    -0.0789 Linea~
##  4 studytime   0.375     0.269      1.39  0.164         -0.154     0.904  Linea~
##  5 absences    0.0374    0.0267     1.40  0.162         -0.0151    0.0900 Linea~
##  6 sexM        1.41      0.451      3.13  0.00188        0.525     2.30   Linea~
##  7 higheryes   1.78      1.03       1.73  0.0837        -0.238     3.80   Linea~
##  8 failures   -0.976     0.252     -3.87  0.000132      -1.47     -0.479  Linea~
##  9 Medu        0.464     0.150      3.08  0.00220        0.168     0.760  Linea~
## 10 goout      -0.462     0.146     -3.17  0.00166       -0.749    -0.175  Linea~
## 11 studytime   0.416     0.200      2.08  0.0381         0.0229    0.809  Linea~
## 12 absences   -0.0658    0.0197    -3.33  0.000955      -0.105    -0.0269 Linea~
## 13 sexM        0.811     0.333      2.44  0.0154         0.156     1.47   Linea~
## 14 higheryes   0.0917    0.862      0.106 0.915         -1.60      1.79   Linea~
## 15 failures    0.420     0.181     -4.81  0.00000153     0.290     0.590  Logis~
## 16 Medu        1.11      0.113      0.900 0.368          0.887     1.38   Logis~
## 17 goout       0.705     0.108     -3.23  0.00125        0.568     0.870  Logis~
## 18 studytime   1.09      0.153      0.584 0.559          0.813     1.48   Logis~
## 19 absences    0.984     0.0137    -1.17  0.244          0.958     1.01   Logis~
```

```
## 20 sexM        1.74      0.256      2.17  0.0300         1.06      2.89   Logis~
## 21 higheryes   2.13      0.561      1.35  0.177          0.715     6.62   Logis~
## 22 failures    0.510     0.192     -3.49  0.000474       0.346     0.739  Logis~
## 23 Medu        1.10      0.128      0.750 0.453          0.855     1.42   Logis~
## 24 goout       0.652     0.124     -3.43  0.000594       0.509     0.830  Logis~
## 25 studytime   1.13      0.176      0.681 0.496          0.803     1.60   Logis~
## 26 absences    0.951     0.0159    -3.14  0.00168        0.920     0.980  Logis~
## 27 sexM        1.49      0.285      1.40  0.162          0.855     2.63   Logis~
## 28 higheryes   1.40      0.651      0.519 0.604          0.378     5.02   Logis~
```

```
write.csv(sensitivity, "results/sensitivity_analysis.csv", row.names = FALSE)
glance(linear_model_nz)
```

```
## # A tibble: 1 x 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.180         0.164  2.95      10.9 1.68e-12     7  -889. 1796. 1831.
## # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Residual plot without the zero group.
ggplot(augment(linear_model_nz), aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.6, colour = "grey45") +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_smooth(se = FALSE, colour = "black", method = "loess") +
  labs(title = "Residuals versus fitted values (students with G3 > 0)",
       x = "Fitted G3", y = "Residual")
```



```r
save_fig("fig9_residuals_nonzero", width = 6, height = 4)
```

Question 6. Draw appropriate conclusions

```r
# No new code. Conclusions are discussed in the report. In short:
#  - Observational, cross-sectional, questionnaire-based data from two
#    Portuguese schools (2005-2006): no causal interpretation.
#  - Previous failures is the most robust and practically important
#    association across all models and both samples.
#  - Absences: the apparent positive or null association in the full data is
#    an artefact of the G3 = 0 group; among students with a final grade,
#    more absences are associated with lower grades.
#  - The models explain only around one fifth of the variation in G3.

# Save session information for reproducibility.
writeLines(capture.output(sessionInfo()), "results/session_info.txt")

################################################################################
# END OF SCRIPT
################################################################################
```